

Yatharth Agarwal

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Summary

Purdue ECE Ph.D. researcher developing safe and efficient robot manipulation systems through VLA policies, optimization-based control, and multimodal perception. Experience in 3D perception, embedded systems, sim-to-real evaluation, and hardware-aware deployment on physical robots and heterogeneous edge computing platforms.

Education

Purdue University *Ph.D., Electrical & Computer Engineering (expected Dec 2027)* **Aug. 2025 - Present**

Advisor: Prof. Vijay Raghunathan *Coursework:* Robotics & IoT, SOC Design, Hardware for ML, Programmable Accelerators

Purdue University *M.S., Electrical & Computer Engineering – GPA 3.75/4.0* **Aug. 2023 - Jul. 2025**

Coursework: AI Foundational Models, MOS & Advanced VLSI, Computer Architecture, Digital Systems Design Automation

Manipal Institute of Technology *B.Tech, Electronics & Communication Engineering – GPA 8.94/10.0* **Jul. 2019 - May 2023**

Research & Engineering Experience

Embedded Systems Laboratory, Graduate Research Assistant *Purdue University* **Oct. 2023 - Present**

- Design and deploy robotic systems spanning VLA manipulation, 3D perception, optimization-based control, and efficient edge inference.
- Build simulation-to-hardware pipelines across MuJoCo, SO-101 manipulators, and heterogeneous CPU/GPU/NPU platforms.

VSDSquadron, Embedded Hardware & Firmware Developer *Remote (Part-time)* **Mar. 2023 - Present**

- Owned PCB/firmware development and BOM optimization with OEMs for RISC-V/FPGA boards; contributed to 5,000+ units shipped.
- Developed C/C++ firmware for board bring-up and hardware diagnostics across SPI, I2C, and UART interfaces on RISC-V/FPGA boards.

Cisco, Hardware Engineering Intern *Bengaluru, India* **Jan. 2023 - Jun. 2023**

- Evaluated next-generation 36-layer router PCBs supporting 10-Gb/s links through DFM/DFA analysis, balancing yield and cost.

Selected Publications & Research

BRACE: Steering VLA Policies Around Moving Hazards with Multi-Link Barriers *Y. Agarwal, et al.* **(Under Review)**

- Developed a training-free multi-link CBF-QP shield for $\pi_{0.5}$ VLA manipulation, combining gripper/wrist/forearm barriers with motion aware hazard tracking; reduced moving-hazard collision from 65.6% to 27.3% and increased safe-success from 29.4% to 50.4%.
- Deployed the perception-control-policy stack on heterogeneous edge hardware and a physical SO-101, achieving 2.2 ms p99 CPU latency for the five-element safety barrier during hardware manipulation.

Edge-Ready 3D Scene-Language Reasoning with Compact Object Graphs *Y. Agarwal, et al.* **IJCNN, 2026**

- Introduced compact object graphs for zero-shot language-guided 3D object grounding and spatial reasoning, achieving 44.5 Acc@0.50 on ScanRefer, 2.6 percentage points above the strongest compared baseline.
- Optimized Jetson AGX Thor inference with GPU superpoint processing, prefix/KV caching, and NVFP4; reduced batch-1 latency from 955 to 298 ms (3.2 $\times$) and peak GPU memory by $\sim$ 33% to 11 GB.

SecuPilot: A Security Coprocessor-Integrated Platform for Autonomous UAV Security *Y. Agarwal, et al.* **ACM TECS, 2025**

- Designed a security-coprocessor PCB and firmware implementing root-of-trust checks, secure boot, firmware verification, and device authentication for PX4/Pixhawk systems.
- Built a hardware-in-the-loop testbed with physical flight controllers and Gazebo dynamics; evaluated benign and adversarial scenarios with > 90% attack-detection accuracy and $\sim$ 11% runtime energy overhead.

Robotics Projects

Autonomous Unmanned Aerial Vehicle *Project MANAS* **May 2021 - Jul. 2022**

- Led a 50+ member cross-disciplinary team developing a 4 kg-payload autonomous UAV with onboard navigation, perception, and tethered payload delivery; coordinated AI, mechanical, electrical, and systems integration.
- Designed 300 A propulsion power-distribution PCBs and low-voltage control electronics; tuned PX4 PID loops for stable flight.

Quadruped CAN Interface & Embedded Bring-Up *Indian Institute of Science* **Jul. 2021 - Sep. 2021**

- Designed a CAN interface shield for STM32-based quadruped leg controllers and supported firmware bring-up for distributed actuator and sensor communication.

Technical Skills

Programming & ML | Python, C/C++, PyTorch, CUDA, Linux, vLLM

Robotics & Control | SO-101/LeRobot, MuJoCo, Gazebo HIL, PX4/Pixhawk, manipulation, CBF-QP, PID control

Perception & Edge AI | VLAs/VLMs, 3D segmentation, RGB-D, LiDAR, NVIDIA Jetson, Intel NPU/OpenVINO, INT8/NVFP4

Embedded & Hardware | STM32/ARM, RISC-V/FPGA; CAN, SPI, I2C, UART, Altium, KiCad, PCB/firmware bring-up